

On the Relationship Between Science, Religion, and Artificial Intelligence from the Perspective of Rationalism

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1 The Concept of the Rational

In philosophy, there are several definitions of rational reasoning, but they all share a common foundation. Rationalism concerns making decisions in such a way that, when choosing among several alternatives, the choice is made according to the following criteria:

- Among several alternatives, the one that is most beneficial is chosen, in the sense of producing results that can later be evaluated objectively.
- Among several alternatives that are equally good according to the first criterion, the simplest one is chosen.
- The existence of conflicting interests among different individuals and interest groups is taken into account through the introduction of game theory.

Ethics represents the common interest of society. It encompasses both norms and the means of their implementation, which are formulated in accordance with the above rationalistic criteria.

Simply put, rationalism comes down to the question “what ought we to do”, understood in terms of criteria that enable us to evaluate outcomes objectively. Objectivity means that the outcome of the evaluation is not open to dispute.

Rationalism is defined by these criteria, that is, by these goals. However, this raises the question of which methods achieve them to the greatest extent.

2 The Concept of Science

Science is defined as a systematic, objectively verifiable, and rational way of acquiring knowledge.

- Systematicity means that there exists a methodology that is applied consistently. Of course, scientific fraud and other problems do occur, but there is also a system for dealing with such problems.
- Verifiability means publishing results in a form that allows anyone to reproduce the procedure, verify the results and the reasoning behind the conclusions, and publish the results of their own verification.
- Objectivity means that the outcome of an evaluation is not open to dispute, that is, there is no room for “subjective judgment”.
- Knowledge is a true belief arrived at through rational reasoning. Belief, on the other hand, is a conviction that lacks a rational foundation. Nevertheless, in this text the term “knowledge” will be used to denote a conviction that has a rational basis, regardless of whether it is actually true or whether it is a misconception.

3 Scientific Misconceptions and Absolute Truth

It is well known that there have been many propositions that were once considered scientifically established but were later refuted by further advances in science. There are also numerous examples in which there was no scientific fraud, the prescribed procedures were followed completely, no procedural errors were made, and yet the conclusions turned out to be far from correct.

Such examples show that scientific misconceptions exist at any given time, and that some of them will be exposed by future scientific progress. In other words, scientific truths are generally not absolute. This means that satisfying the conditions required for accepting a scientific proof is not a complete guarantee that the proposition being established is true.

However, such a relativization of science has its limits. It obviously makes no sense to claim that there is a small possibility that the Earth is flat. Taking into account the existence of scientific misconceptions, it is possible to establish that certain scientific truths are free of such misconceptions. In philosophy, this is known as the problem of absolute truth.

Absolute proofs are not possible in the logical sense. Every proof is based on certain assumptions, which may themselves be methodological. If the level

of skepticism is raised too high, then nothing can be proven—not even that the monitor screen on which this text is being read is closer in shape to a rectangle than to an ellipse. Suppose we doubt both our senses and our reason. What, then, shall we rely on? Much weaker assumptions are sufficient to make it impossible to prove even the most obvious facts.

However, in the empirical sense, absolute proofs may exist in the form of filters that admit certain truths while excluding all falsehoods. Of course, the property of such a filter that it excludes all falsehoods could not itself be proven. It could only be accepted, though not blindly, but on the basis of very strong intuitions that it is unreasonable to challenge such principles of proof.

The subject of absolute truth is deep and complex and therefore cannot be treated in detail here. Naturally, one may immediately ask why scientific methodology is not adjusted so as to eliminate scientific misconceptions altogether. The problem is that such an approach would yield far fewer results, and science would be much less useful.

Science, together with its methodology, is an integral part of rationalistic philosophy. This means that the methodology accepted today is a more rational choice than a more restrictive methodology that would eliminate scientific misconceptions. In this way, rationalistic philosophy establishes the principle of utility as the criterion with which scientific methodology is aligned.

The methodology of every science is aligned with the goal of enabling that science to function as effectively as possible, producing as many results as possible while ensuring that their application does not lead to problems, although guarantees of this kind are generally incomplete.

In other words, scientific methodology is aligned with the principle that, when scientific results are applied at the object level of the sciences concerned, it is most useful to accept those results as a whole while disregarding the existence of scientific misconceptions.

However, there are considerations at the meta level in which the existence of scientific misconceptions plays a role. Of course, the meta level of one field is the object level of another field, which has its own methodology aligned with rationalistic criteria.

3.1 Is Scientific Methodology Universally Applicable?

First, not all sciences employ the same methodology. For example, in the natural sciences, experimentation is accepted as evidence, whereas this is not the case in mathematics. Nevertheless, the natural sciences constitute a broad family of disciplines and share a common methodology. In many contexts, the term *scientific methodology* is in fact used to refer specifically to the methodology of the natural sciences. Why, then, is this methodology not applicable to mathematics?

Mathematics has historically developed in a deductive manner and has reached a level involving such delicate concepts and propositions that even small logical errors lead to contradictions. If mathematicians were to accept experimentation as evidence in mathematics, this would result in a large number of mutually contradictory propositions, with no way to determine which should be accepted and which rejected, and mathematics would cease to function. One might even say that the current level of rigor in mathematical exposition is the lowest level of rigor at which modern mathematics can still function.

It turns out that mathematics is an undisputed example of a field in which the methodology of the natural sciences is not applicable. However, this means that the methodology of the natural sciences is not universally applicable, and therefore there may be other fields in which it is likewise inapplicable.

This means that it is a mistake to apply the methodology of the natural sciences uncritically outside the domains in which it has proven successful. Unfortunately, this mistake is often made. For example, there are contexts in which the meta level of the natural sciences becomes relevant. In such cases, it is a mistake not to engage in the necessary meta-level considerations. Rationalistic philosophy provides a framework for doing so.

Let us now consider some deeper explanations of the empirical fact that the methodology of the natural sciences is applicable across a broad family of natural sciences, but not in mathematics.

3.2 Why Is the Methodology of the Natural Sciences Applicable in the Natural Sciences?

The natural sciences deal with natural phenomena, which can ultimately be reduced to the fundamental laws of nature studied by physics. Here we restrict our attention to phenomena within our own Big Bang. According to the current understanding of the nature of fundamental laws (and consequently of derived laws), the concepts of probability and algorithm are embedded in the foundations of the laws of nature in the following sense:

The state of the present uniquely determines the probabilities of future events. Moreover, for any prescribed level of accuracy, it is possible to compute algorithmically the probability of a given future event to that accuracy, provided that sufficiently accurate information about the present is available and that sufficient computational resources are available to carry out the calculation. The equations of physics are such that they permit numerical simulations.

Of course, existing physical theories represent only our current understanding of the laws of nature, which is partial and approximate. Therefore, the character of the laws that actually govern nature cannot be inferred solely from the character

of the laws embodied in currently accepted physical theories. The assumption that the laws governing nature possess the character described above is known as the Church–Turing–Deutsch thesis.

It follows from this thesis that it is impossible to construct a machine that would solve a non-computable problem with a reliability greater than, say, 90%. Indeed, if a machine solves a problem with a reliability greater than 90%, then for any given input values we could algorithmically determine, by simulating the operation of that machine, which output is produced with a probability greater than 89%. That output would then constitute a solution to the problem, implying that the problem is algorithmically solvable.

From the assumption that these conditions hold in a given universe, it follows that the use of experimentation and Occam’s razor (the principle according to which, among all explanations of a given set of phenomena, the correct one is the simplest) in the study of phenomena within that universe is justified. This explains the applicability of these methods in the natural sciences.

The converse also holds. The methodology of the natural sciences is applicable precisely in those domains in which the Church–Turing–Deutsch thesis holds and in which phenomena are measurable in principle. In this text, we shall use the term *natural phenomena* to denote phenomena that are measurable in principle and that are subject to the Church–Turing–Deutsch thesis.

This is consistent with the need to describe phenomena in a way that allows the prediction of the outcomes of processes and, consequently, the practical applicability of the sciences that study them.

3.3 Why Is the Methodology of the Natural Sciences Not Applicable to Mathematics?

If the natural sciences concern nature and the laws governing natural phenomena, then what does mathematics concern? This question is addressed by the philosophy of mathematics, within which several schools of thought exist. The most widely accepted among them is mathematical Platonism, in its various forms, which emerged during the twentieth century and was inspired by Plato’s allegory of the cave.

According to traditional, or pure, Platonism, mathematics concerns an objective reality in the form of a kind of mathematical universe. In the full-blooded version of Platonism, however, the mathematical universe is replaced by a mathematical multiverse containing all consistent combinations of axioms and all possible varieties of logic built upon them.

Mathematical Platonism is the principal form of mathematical realism (as opposed to mathematical anti-realism). Some hold the view that mathematical re-

alism is the only remaining possibility. In any case, mathematical Platonism is incompatible with the classical form of materialism.

Consider a computer (equipped with infinite memory) in which every program accepts a file named “**input**” as its sole input and produces a file named “**output**” as its sole output. Given a program and its input file, we may ask the following question: Will that program, when run on that particular input, terminate after a finite number of steps? Recall that programs may contain infinite loops.

We may also ask the following related question: Is it possible to write a program *H* such that, whenever its input file is a ZIP archive containing two files named “**program**” and “**in**”, the following conditions hold:

- Program *H* always terminates after a finite number of steps.
- Upon termination, the file “**output**” contains the correct answer, “yes” or “no”, to the following question:

*Does the program stored in the file “**program**”, when executed with an input file having the same contents as the file “**in**”, terminate after a finite number of steps?*

We call this the halting problem. Such a program *H* cannot be written. To prove this, let us derive a contradiction from the assumption that such a program *H* exists. Under this assumption, it is possible to write a program *Q* that performs the following steps on a given input file:

- It creates a ZIP archive containing two files named “**program**” and “**in**”, each of which is a copy of the file provided as input to program *Q*.
- By simulating the execution of program *H* on this ZIP archive as its input file, it obtains *H*’s output.
- If the answer obtained is “yes”, then program *Q* enters an infinite loop. If the answer is “no”, then program *Q* terminates.

Before running a program *R*, we may place any content we wish into its input file. For example, we may place the source code of program *R* itself into the input file. We may then execute program *R* with this input file.

For every program *R*, the following holds. When program *R* is executed with an input file containing the description of program *R* itself, then:

- If program *R* terminates after a finite number of steps, then program *Q*, when executed with the same input file, never terminates.
- If program *R* never terminates, then program *Q*, when executed with the same input file, terminates after a finite number of steps.

Since this holds for every program R , it must also hold when R is program Q itself. If we execute program Q with an input file containing the description of program Q , then program Q terminates after a finite number of steps if and only if it never terminates. This is, of course, a contradiction.

This means that the halting problem cannot be solved algorithmically. Here the word “problem” refers to the production of the answer “yes” or “no” for an arbitrary input file in a deterministic sense: no matter how many times the procedure is applied to the same input file, it always produces the same output.

By an ordinary computer we mean a computer that can solve every algorithmically solvable problem and nothing more.

Let us imagine a computer C that differs from the computer described above in that it has been augmented with an operation that need not be algorithmically computable, but is deterministic in the sense defined above. We shall call such an operation a special operation.

We may consider the halting problem for computer C and ask whether there exists a program for computer C that solves the halting problem for computer C . The answer is “no”, by an argument analogous to the one given above.

Of interest here is the computational power of such a computer in the following sense: For which tasks is it possible to write a program for that computer? If the special operation performs something that is algorithmically computable, then the computational power of that computer, in this sense, is the same as that of an ordinary computer.

Suppose that M is a computer augmented with a special operation S that solves some particular problem. We may then consider a computer N augmented with an operation F that solves the following problem. The input consists of the contents of a ZIP archive containing two files named “**what**” and “**data**”. If the file “**what**” is empty, then operation F produces the output of operation S on the input data contained in the file “**data**”. If the file “**what**” is not empty, then operation F solves the halting problem for computer M , where the input is the contents of the file “**data**” (which is itself a ZIP archive containing the files “**program**” and “**data**”).

This computer can solve every problem that computer M can solve, but it can also solve the halting problem for computer M , which is unsolvable for computer M . We call this computer N the successor computer of computer M .

Consider an infinite sequence of computers $M_0, M_1, M_2, \dots$ where M_0 is an ordinary computer, M_1 is the successor computer of M_0 , M_2 is the successor computer of M_1 , and so on. Each of these computers is mathematically well defined, as is its corresponding halting problem.

We cannot build any of these computers, but we can write programs for any of them. Programs are written in the usual way, with the special operation S

denoted by some chosen notation. This means that the halting problem for any of these computers is an entirely mathematical problem.

Consider the general halting problem formulated as follows. The input consists of a ZIP archive containing the index of a computer from the infinite sequence described above, a program for that computer, and an input file. The question is whether that program, running on that computer with that input file, terminates on that input. This problem is entirely mathematical, yet it cannot be solved by a program running on any of the computers in the infinite sequence described above.

Consider the simulation of a stochastic process on one of the computers in the above sequence, where the input file contains a large finite sequence of zeros and ones that is used as a sequence of random bits. The program must answer “yes” or “no” to the following question: Did a particular random event occur during the simulation or not? Sometimes it will be impossible to determine whether the event occurred, because all the bits from the input file have been consumed during the simulation while additional bits are still required. In that case, the answer will be “maybe”.

The answer depends on the sequence of zeros and ones provided in the input file. We may choose some large length n of the finite bit string and perform the simulation once for each bit string of length n .

We shall then define the lower bound of the probability of the random event under consideration as the ratio of the number of bit strings for which the answer is “yes” to the total number of bit strings, which is 2^n . Similarly, we shall define the upper bound of the probability of the random event as the ratio of the number of bit strings for which the answer is either “yes” or “maybe” to the total number of bit strings.

In this way, we have associated with each value of n a lower and an upper bound for the probability of the event.

When n is very large, the difference between the lower and upper probability bounds is small, and the two bounds converge to the same value in the limit as n tends to infinity. This allows us to define the probability of the random event under consideration as the common limit of the lower and upper bounds as the length n of the finite bit string tends to infinity.

This procedure does not define the probability of every event in probability theory, because the sequences of bounds described above do not always converge to the same value. Whenever the probability is defined in this manner, however, the resulting value is correct.

Nevertheless, this procedure does define the probabilities of events that are relevant to the natural sciences, because it is consistent with the way natural phenomena are simulated on a computer. More precisely, for all measurements that are actually performed, the relevant events are Jordan measurable, and for

such events probability is defined in the manner described above.

Consider the assertion that a given rational number is greater than (or less than) the probability of a random event associated with a stochastic process that can be simulated on an ordinary computer. Such assertions, for a given rational number, are decidable by some computer in the infinite sequence described above. However, no computer in that sequence can solve the general halting problem, even though it is a mathematical problem.

Therefore, the character of mathematical laws does not satisfy the Church–Turing–Deutsch thesis, which means that the character of mathematical laws is fundamentally different from that of natural laws. In light of this, it is not surprising that the methodology of the natural sciences is not applicable to mathematics.

There are also examples of mathematical results that could not have been obtained empirically. One such example is the classification theorem for finite simple groups. There are several infinite families of finite simple groups, together with 26 additional finite simple groups (known as the sporadic finite simple groups) that do not belong to any of those families.

The classification theorem states that these are all finite simple groups. The number of elements in the largest sporadic finite simple group is

$$808017424794512875886459904961710757005754368000000000.$$

The description of some of these groups is so intricate that they could not have been discovered empirically. The way they were discovered was by attempting to prove that no finite simple groups exist other than those that were known at the time.

If those known groups did not in fact constitute the complete list of finite simple groups, then such a proof could not succeed. This manifested itself as a particular obstacle encountered in the attempt to prove the statement. By analyzing that obstacle, mathematicians were led to finite simple groups that had previously been unknown.

If, by chance, one were to rely on the false assertion that only a subset of these finite simple groups constituted the complete collection of finite simple groups, many incorrect conclusions would be derived from it within group theory. Eventually, this would lead to mutually contradictory statements, with no way of determining which should be retained and which should be discarded.

3.4 The Limits of the Relativization of Science

3.4.1 Criteria

The problem of absolute truth is profound and cannot be examined in detail here. Instead, we shall rely on simpler intuitions.

Despite the existence of scientific misconceptions, we may reasonably maintain that there is no genuine possibility that the Earth is closer in shape to a cube than to a sphere. Even if humanity were to alter the shape of the Earth through future technologies, the historical fact that, in our time, the Earth was more similar in shape to a sphere than to a cube would remain an absolute truth. In other words, we are not mistaken about this.

One might say that, in this case, we are dealing with measurable quantities based on scientific theories that may not be absolutely correct, but which cannot be so inaccurate as to imply that the Earth is closer in shape to a cube than to a sphere. Indeed, these theories are being applied to the Earth under conditions in which they have been thoroughly tested and confirmed with great precision.

However, it is important to take into account the fact that some things once believed to be immutable were later shown to be subject to change. For example, prior to the twentieth century, it was believed that the continents were fixed and unchanging. Today, it is known that this is not the case.

In addition, it is important to note that the fact that natural phenomena are governed by certain laws does not exclude the possibility of the existence of God. We may imagine a scenario in which God created the world through the Big Bang, establishing its laws, constants, and parameters, and then allowed it to operate according to those principles. Such a formulation of religion is not in conflict with science. However, one should be cautious with respect to the two metaphors that we shall discuss below.

3.4.2 Darwin's Theory of Evolution

Darwin's theory of evolution is a scientific theory that predicts and describes the evolution of life forms (without explaining the origin of life) as a completely natural process governed by the following three Darwinian principles:

- the principle of natural selection,
- the principle of inheritance of traits,
- the principle of random mutation.

There is a large body of scientific evidence supporting Darwin's theory of evolution. It is also interconnected with other biological phenomena. It is generally held that all biological phenomena must be interpreted in evolutionary terms in order to be properly understood.

However, the same could have been said of Newtonian mechanics in the year 1900. At that historical moment, the following statements could have been made about Newtonian mechanics:

- Newtonian mechanics had been tested by numerous experiments with great precision, and the theory had always been consistent with experimental results within the limits of measurement accuracy.
- Newtonian mechanics possessed a mathematical model demonstrating that it contained no internal contradictions, that is, that it could not be refuted purely through mathematical reasoning.
- Newtonian mechanics had been cross-validated against many phenomena from other fields, and no problems had been found.
- Many natural phenomena had been successfully explained by means of Newtonian mechanics.
- Nothing had been observed that suggested Newtonian mechanics was not an absolute truth.
- This favorable state of affairs had persisted for more than two hundred years.

Approximately the same can be said about Darwin's theory of evolution today. Many biological phenomena, both those occurring in the present and those relating to the history of life forms, have been explained by Darwin's theory of evolution. Many evolutionary transitions have been explained by it; some have not yet been fully explained, but nothing has been observed that would be inconsistent with the possibility that Darwin's theory of evolution is completely correct.

However, at the beginning of the twentieth century it was demonstrated that Newtonian mechanics is not entirely correct. This historical experience teaches us that the facts listed above are not sufficient to rule out the possibility that Darwin's theory of evolution may not be completely correct.

Nevertheless, contemporary scientific understanding of biological evolution cannot be mistaken to the extent required to challenge the evolutionary origin of human beings from much simpler forms of life. Quite simply, there exist fossil remains of transitional life forms that conclusively exclude such a possibility.

Thus, it is conceivable that some future Einstein of biology may develop a more advanced theory of biological evolution, leading biologists to say that Darwin's theory of evolution has only limited applicability, that Einstein's theory of evolution has broader applicability, and that our understanding of life has changed. Yet even under such a more advanced theory, human beings would still have an evolutionary origin from incomparably simpler forms of life, because that is a fact.

3.4.3 The Origin of Life

Although Darwin's theory of evolution provides an excellent account of the transformation of life forms, it does not explain the origin of life. The process by which life arises from nonliving matter, as well as the scientific field that studies this process, is known as abiogenesis.

It is possible that life did not originate on Earth but rather on some other planet and was subsequently transferred to Earth. However, this does not explain the origin of life; it merely relocates it to another planet. Therefore, we shall assume here that life as we know it originated on Earth.

Darwinian principles are not fully applicable to the period when the Earth was young and when a self-replicating molecule with the necessary properties had not yet been synthesized.

Abiogenesis is the process that leads to the fulfillment of the minimum conditions required for Darwinian evolution, and this is precisely what connects it to Darwin's theory of evolution. Furthermore, Darwinian principles are partially applicable during certain stages of abiogenesis.

Today, abiogenesis is generally regarded as a completely natural process, primarily involving chemistry and the geological evolution of the early Earth. However, not all stages of abiogenesis have yet been explained. Abiogenesis gave rise to the simplest unicellular organisms, whose functioning is in any case entirely reducible to chemistry and is therefore natural.

Nevertheless, a selective process is required for the transition from nonliving matter (the primordial soup) to something capable of Darwinian evolution. If such a system were to arise through a single random event, that event would be so improbable that the probability of its occurring at least once somewhere within our Big Bang, during a period no longer than the age of our Big Bang, would be negligible.

Imagine that we need to develop a commercially competitive word processor. One possible approach would be to generate a sufficiently long random sequence of bytes, save the resulting file under the name `setup.exe`, and hope that we were fortunate enough to obtain a fully functional installation wizard for a powerful and reliable word processor. In order for the probability of success to become significant, an enormous number of attempts would be required.

Similarly, if abiogenesis depended on a single random event, then achieving a significant probability of success somewhere within our Big Bang would require so many attempts that the matter cannot be explained merely by multiplying the number of galaxies, the number of planets per galaxy, the amount of water in seas and oceans, and the age of the Big Bang.

One explanation is intelligent design, namely that the first life form was created by some powerful intelligent being. This explanation could be challenged by

demonstrating that the first life form was far from perfect.

A second explanation is a selective process, which is not yet fully understood, although some of its components are known. Support for this explanation is sought both by identifying mechanisms through which abiogenesis could have occurred and by discovering even the simplest forms of life elsewhere.

The planet K2-18b has been reported to contain dimethyl sulfide, a substance that is believed to exist on Earth exclusively as a consequence of life. However, both the detection of dimethyl sulfide and the claim that it cannot arise in the absence of life remain controversial.

The explanation based on a selective process could be challenged if invariants were discovered that cannot be violated and that prevent the emergence of the first life form.

A third explanation is that our Big Bang is one of infinitely many Big Bangs within a larger cosmos, so that no matter how small the probability of success may be, infinitely many successes will occur given infinitely many attempts.

However, this would imply that we happen to exist in one of those exceptionally rare Big Bangs. This is not surprising, since we could not have appeared in a different kind of Big Bang. Yet among the Big Bangs in which intelligent life forms capable of discussing such questions arise, those in which abiogenesis occurred more than once would be exceedingly rare. Consequently, such an explanation would be refuted if life were discovered elsewhere in our galaxy.

Therefore, especially if we are concerned with absolute truth rather than merely scientific truth, no reliable position can currently be taken regarding abiogenesis, not even concerning its general character.

3.4.4 Why Do We Seek Scientific Explanations?

We could have attributed thunder to some sort of supernatural beings. Later, however, a scientific explanation was discovered, which ultimately made electrical technologies possible. In this way, the discovery of a scientific explanation proved useful, because such phenomena can then be brought under control.

The possibility cannot be ruled out that there exist phenomena that do not satisfy the Church–Turing–Deutsch thesis, that cannot be explained by the methodology of the natural sciences, and that cannot be brought under control. However, there is no way to prove that a particular phenomenon satisfies these conditions. Consequently, abandoning the search for scientific explanations of phenomena that have not yet been scientifically explained would amount to surrendering without a struggle.

3.4.5 What Can We Claim to Be Absolute Truth?

According to the criteria outlined above, we may claim as an absolute truth that humans evolved from ape-like ancestors and that humans and crocodiles share common ancestors.

On the other hand, there is a genuine possibility that biology may undergo a revolution similar to the one that occurred in physics—that some biologist, playing a role analogous to Einstein’s in physics, may develop a more advanced theory of biological evolution (and perhaps of other phenomena as well), and that it may then be discovered that Darwin’s theory of evolution is applicable only under certain conditions, much as Newtonian mechanics is.

Nevertheless, many phenomena have been successfully explained by Darwin’s theory of evolution, and many of those explanations can be identified as indisputable according to the criteria discussed above, which means that there is no genuine possibility that they are incorrect.

This means that Darwin’s theory of evolution, as a theoretical model of biological evolution, is at least approximately correct under certain conditions, much like Newtonian mechanics. However, it is important to consider a problem that can be illustrated by the example of gravitational theory.

General relativity provides a more advanced description of gravity than Newton’s theory of gravitation. Under ordinary terrestrial conditions, the two theories yield observationally similar descriptions of gravity. In such circumstances, the differences in their predictions are so small that they are extremely difficult to measure. Nevertheless, the two theories are conceptually very different.

General relativity is generally believed not to be completely true, primarily because of its incompatibility with quantum phenomena. Yet its description of gravity is certainly more accurate than that of Newton’s theory of gravitation. Moreover, all effects predicted by general relativity have been confirmed empirically, and no empirical deviations from the theory have been observed.

According to general relativity, gravity is entirely a consequence of the curvature of spacetime. This curvature is so great that it is difficult to overcome. If we jump upward, we very quickly fall back to the ground. Sending something into space beyond Earth’s orbit requires a large amount of energy, which makes it very expensive.

According to Newton’s theory, on the other hand, spacetime curvature does not exist at all, and gravitational effects are attributed to a gravitational acceleration field whose origin is attributed to mass, although the mechanism behind this is not explained within the theory. The theory nevertheless provides a mathematical description of that acceleration field, making calculations possible.

General relativity predicts a degree of spacetime curvature in ordinary terrestrial conditions that is sufficiently great to make overcoming it difficult.

Yet the two theories yield approximately the same observable predictions in weak gravitational fields. Earth's gravitational field is sufficiently weak that gravity alone cannot accelerate us to speeds comparable to the speed of light, where relativistic effects become significant.

Accordingly, it is possible that in some more advanced theory of biological evolution, biological evolution will be explained in a completely different manner, perhaps even by effects that are currently regarded as nonexistent (just as space-time curvature does not exist according to Newton's theory of gravitation), while the applicability of Darwinian principles under certain conditions would emerge as a consequence of those other principles.

Today, biological evolution is regarded as a completely natural process because no evidence to the contrary has been observed. This assessment could potentially change in the future within the following framework:

- that some currently unknown factors influenced certain phenomena that remain unexplained at present,
- that the applicability of Darwinian principles (at least under certain conditions) follows from something that is currently regarded as nonexistent,
- that there exists something at the meta level that is not encompassed by current understandings of evolution.

One possibility under which biological evolution would not be a completely natural process is that consciousness is a metaphysical category, that possessing or using consciousness requires a suitable natural organ (such as the brain) with which it interacts, and that natural selection guided the development of the brain toward increasingly effective utilization of consciousness.

4 Why Is the Argument Against the Existence of God Based on Omnipotence Incorrect?

If we assume that an omnipotent God exists, then such a being can create a stone so heavy that it cannot lift it. But then that being is not omnipotent, since it cannot lift that stone. This is often presented as a proof that an omnipotent God does not exist, because the assumption of His existence leads to a contradiction. However, is this reasoning correct?

If we were to accept this type of argument against the existence of God, then we should also accept similar arguments in other contexts. Here we shall demonstrate the inadequacy of such reasoning by applying it to derive a contradiction within the theory of natural numbers. This derivation is known as Richard's paradox.

A Richard number is defined as a natural number that can be uniquely specified in the Serbian language using no more than 300 characters, including spaces, punctuation marks, and special characters.

An example of a Richard number is 73, because the Serbian language includes the decimal numeral system, and “73” is a unique description of that number in Serbian consisting of only two characters. Similarly, every number whose decimal representation contains fewer than 300 digits is a Richard number. However, the number 10^{500} requires more than 300 digits in decimal notation, yet it is a Richard number because it can be uniquely described in Serbian by the expression “10 to the power of 500”, which contains far fewer than 300 characters.

There are many Richard numbers, but there are only finitely many of them. The Serbian alphabet contains 30 lowercase letters, 30 uppercase letters, 10 digits, a space character, punctuation marks, and special characters. Let us suppose that the total number of characters in the Serbian alphabet is 100. Then the total number of character strings of length, for example, 27 is equal to 100^{27} . Therefore, the total number of character strings of length at most 300 is

$$100^1 + 100^2 + \cdots + 100^{300}.$$

This number is large, but finite. Let us denote the total number of character strings of length at most 300 by N . Only some of these character strings are meaningful in Serbian and uniquely determine a natural number. Therefore, the number of meaningful Serbian character strings of length at most 300 that uniquely determine a natural number cannot exceed N .

The same natural number can often be defined in Serbian in more than one way using no more than 300 characters. For example, the number 56 can be written as “56”, but also as “50 plus 6”. Consequently, there are even fewer Richard numbers, and in any case there cannot be more than N of them.

There are $N + 1$ natural numbers from 1 to $N + 1$, which is more than N , and therefore not all of them can be Richard numbers. Hence, there exist natural numbers that are not Richard numbers. Moreover, at least one such number is not greater than $N + 1$.

There is also a smallest natural number that is not a Richard number. Clearly, there cannot be two natural numbers each having the property of being the smallest natural number that is not a Richard number, since one of them would have to be smaller than the other. Therefore, the expression

“The smallest natural number that cannot be defined in the Serbian language using no more than 300 characters, including spaces, punctuation marks, and special characters.”

uniquely determines the smallest natural number that is not a Richard number. Since the expression above contains fewer than 300 characters, we arrive at the

conclusion that the smallest natural number that is not a Richard number has the property of being a Richard number.

This is a contradiction, because the same number cannot simultaneously be a Richard number and not be a Richard number. Since natural numbers do exist, this derivation of a contradiction cannot be correct. What we have here is a paradox of language: the problem is real, but analysis of the paradox leads us to conclude that everything else is in order.

This means that natural languages permit constructions that are linguistically well formed when meaning is ignored, but that in fact have no meaning. Moreover, this cannot be established in a simple way; a deeper philosophical analysis is required in order to determine that the text is meaningless.

This paradox is resolved by formalizing the language. An artificial language is constructed for use within a particular domain, one that does not suffer from the problem described above.

Such a language must be sufficiently expressive for use in that domain, and every linguistic construction in it must have exactly one meaning, with that meaning being explicitly known. In such a language, paradoxes of this kind cannot be formulated.

Richard's paradox is resolved in mathematics precisely in this way. A similar approach is taken in the philosophy of religion, where an analytical framework (with a strong reliance on formal logic) is employed in order to avoid problems arising from improper use of the concepts of omnipotence and omniscience.

5 The Burden of Proof and the Invisible Dragon

There is a view according to which those who claim that God does not exist are not required to prove their claim. Rather, the burden of proof rests solely on those who claim that God exists. In other words, the absence of available evidence for the existence of God is regarded as sufficient grounds for concluding that God does not exist.

This position is often supported by the metaphor of the invisible dragon. If someone were to claim that an invisible dragon lives in their garage, but that this cannot be proven because the dragon is invisible and hides itself skillfully, we might reasonably conclude that such a dragon does not in fact exist.

Let us recall that rational reasoning has been defined as reasoning that leads to useful decisions. The belief that the invisible dragon exists provides us with no advantage over the belief that the invisible dragon does not exist. Indeed, the belief that the invisible dragon exists would itself constitute a burden of some kind. Therefore, the rational position is that the invisible dragon does not exist.

However, such an argument cannot be transferred directly to the question of

the existence of God, because religion typically maintains that, with respect to what happens after our earthly death, it is not irrelevant what position we have taken regarding the existence of God, God's expectations of us, and the fulfillment of those expectations.

One may therefore ask whether the argument involving the invisible dragon can be applied to the question of the existence of God if we reformulate the problem in such a way that the invisible dragon (assuming it exists) becomes important to us in a similar manner.

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There is another metaphor in which someone rings our apartment doorbell and, when we open the door, we see a person with whom the conversation proceeds roughly as follows:

- Would you like to receive one million dollars (a metaphor for the reward after earthly death spoken of by religion)?

- Of course. What do I have to do?
- Leave this city (which serves as a metaphor for earthly death) and go to a particular place, where you will receive one million dollars.
- Has anyone already received one million dollars in that way?
- Certainly.
- Have they contacted you?
- No, because once you go to the place where the million dollars are distributed, you can no longer return to the city or communicate with anyone in the city (which serves as a metaphor for the inability of the dead to communicate with us).
- Have you yourself received one million dollars in that way?
- No. I have not yet left the city, but I certainly intend to do so.
- Then how do you know that anyone has received one million dollars in that way?

The point here is that even if God is presented as being important to us, one can invent all sorts of stories that are not verifiable in any way and therefore amount to complete arbitrariness, and consequently to nothing more than an unnecessary burden.

Furthermore, the kinds of outcomes after earthly death discussed by religion are not objectively verifiable while we are still alive on Earth.

In this way, religious formulations that speak exclusively about the afterlife, without providing any earthly benefit to their followers and without offering any form of verifiability, are deprived of their force.

Nevertheless, many people believe that they have had religious experiences in which God has helped them in some way in their lives. The same could not be said of the invisible dragon.

Let us consider some mathematical examples against the argument that the burden of proof rests only on those who claim that something exists, and not also on those who claim that it does not exist.

Mathematics is founded formally in the sense that there exist algorithmic criteria for determining whether a given string of symbols constitutes a valid mathematical statement and whether a given string of symbols constitutes a valid proof of a statement, and if so, of which statement.

According to Rosser's incompleteness theorem (which is related to Gödel's first incompleteness theorem), any such system is necessarily incomplete in the sense that there exist mathematical statements that are valid expressions in the system but can neither be proved nor disproved according to the rules of that system.

For this reason, there exists the standard axiomatic system *ZFC* set theory (which is capable of serving as a foundation for mathematics), describing the generally accepted forms of mathematical reasoning.

The greatest success of this system is that it is natural in the sense that mathematicians who are unfamiliar with it nevertheless tend to follow it consistently. This may be viewed as a measure of how well the system captures mathematical reasoning.

For the following two statements,

- There exists a nonconstructible set.
- There exists a Suslin tree.

it has been proved that neither can be proved nor disproved from the axioms of *ZFC*, while it also follows from the axioms of *ZFC* that at least one of them must be true. Thus, the issue is not merely that proofs of these statements are currently unknown; rather, it has been proved that neither statement can be proved nor refuted from the axioms of *ZFC*.

If we were to conclude that neither nonconstructible sets nor Suslin trees exist simply because their existence cannot be proved, and if we were to extend the axiomatic system *ZFC* by adding as axioms that nonconstructible sets do not exist and that Suslin trees do not exist, we would obtain an inconsistent system, since it is provable from the axioms of *ZFC* that at least one of these two objects must exist.

Furthermore, suppose we were to adopt the mathematical principle that whenever a mathematical statement asserting the existence of something cannot be proved using certain predetermined (computationally specified) proof resources (which need not be the same for all such statements, provided that the resource bound is algorithmically computable from the statement itself), we then accept the nonexistence of the object in question as an additional axiom. In that case, we would necessarily arrive at a contradiction.

Indeed, suppose that all mathematical formulas asserting the existence of something are assigned natural numbers according to some algorithmic rule, that is, arranged into an infinite sequence F_n (which is entirely feasible). We may then define an infinite sequence of theories as follows. Let T_0 be the theory *ZFC*. We define the theory T_1 as follows:

- If the statement F_1 is provable in the theory T_0 using the specified bounded resources, then T_1 is defined to be the theory consisting of the axioms of T_0 together with the statement F_1 as an additional axiom.
- If the statement F_1 is not provable in the theory T_0 using the specified bounded resources, then T_1 is defined to be the theory consisting of the axioms of T_0 together with the negation of F_1 as an additional axiom.

Similarly, we define the theory T_2 as follows:

- If the statement F_2 is provable in the theory T_1 using the specified bounded resources, then T_2 is defined to be the theory consisting of the axioms of T_1 together with the statement F_2 as an additional axiom.
- If the statement F_2 is not provable in the theory T_1 using the specified bounded resources, then T_2 is defined to be the theory consisting of the axioms of T_1 together with the negation of F_2 as an additional axiom.

The construction of the subsequent theories in the sequence is continued in the same manner. The theory T consisting of all axioms belonging to all theories in the sequence would be complete (every mathematically well-formed statement expressible in the theory would be either provable or refutable in T). Since it would also satisfy the assumptions of Rosser's incompleteness theorem, it would necessarily be inconsistent.

Since every proof is a finite sequence of formulas and therefore uses only finitely many axioms, it would follow that for some natural number n , the theory T_n is already inconsistent. In other words, after finitely many steps we would arrive at an inconsistent system, and consequently at false conclusions and an unusable theory.

7 What Should Religion Look Like?

Given the problems illustrated by the two metaphors discussed above, any formulation of religion that is to be meaningful should possess some form of confirmation concerning the afterlife that is, in principle, verifiable.

Religions often appeal to certain recurring miracles that anyone can allegedly witness for themselves. However, this immediately raises the question of why such phenomena do not lead to mass conversion to the religion in question. Clearly, such claims about verifiable and recurring miracles are not true.

However, another kind of confirmation is possible—namely, that the religion brings some benefit either to society as a whole or to the individuals who embrace it. A social benefit might consist in members of society becoming more honest, conscientious, and compassionate, thereby allowing society to function more effectively.

Yet if every benefit provided by that religion can be replaced to an equal or greater extent by secular measures that would emerge through social development, then that religion would amount to nothing more than a temporarily useful misconception.

If the benefit of the religion consisted solely in creating bonds among its followers, then it would have a secular counterpart in people uniting around something

else. Such a religion would not differ substantially from a mafia organization whose members benefit from belonging to it.

The ideal case would be one in which the followers of a religion benefit from it, while other constructive members of society also benefit from its existence, albeit to a lesser degree.

With regard to the benefits that believers derive from religion, some believers report religious experiences in which God has helped them in their lives.

What should we expect from God's activity in the world? Scientific theories predict probabilities that have a statistical meaning. If all of those probabilities were realized exactly as predicted in reality, then a necessary statistical consequence would be that rationalists are the most successful people. One should expect from God only minor interventions (having no significant impact on other matters) when it is necessary to help a person overcome evil, together with non-interference in everything else.

Why would God intervene in scientific experiments and thereby hinder scientists in their search for truth about the world, when they are engaged in something good and useful? Why would He interfere with technologies created through human effort, when such technologies represent human achievements and when such interference could endanger passengers on an airplane? If God is benevolent toward humanity, then He rejoices in human good deeds.

8 Cosmology and the Problem of Existence

The problem of existence is the search for an answer to the question of why there is something rather than nothing. Introducing the assumption that God exists does not solve this problem, because it does not answer the question of why the God who created everything else exists. The difficulty lies in the transition from nothing to something.

There is also the opposite problem: the transition from something to nothing. Earthly death is often associated with this issue. How does consciousness disappear? What does it become?

In contemporary cosmology, the question of the origin of the Big Bang is addressed by constructing an appropriate mathematical model that requires nothing external to itself, but instead contains within itself everything necessary for its operation. However, such a mathematical model is made of mathematical substance rather than nothing.

Even without assuming mathematical Platonism, every major school within the philosophy of mathematics holds that mathematical laws are about something rather than nothing.

For example, logicism is the view that mathematics can be reduced to logic.

In its basic form, logicism holds that mathematics studies the logical relationships among axioms and definitions. However, in that case, axioms, definitions, and the relationships among them must in some sense exist.

Formalism is the view in the philosophy of mathematics according to which mathematics is the study of formal systems. But then formal systems, together with their properties and the relationships among them, must exist.

Such investigations take into account the fact that physical laws make sense only if they are laws about something. There must be a space to which states correspond, as well as changes in those states. However, something similar also holds for mathematical laws. Consequently, such considerations move one step further back beyond the Big Bang, but not all the way to the end of the problem.

Observations made with the James Webb Space Telescope have led to the rejection of certain cosmological views that were previously considered established. However, this does not alter the conclusions presented above.

9 Problems with Materialism and Reductionism

A natural question arises: why not simply accept materialism? Has anything been observed that poses a problem for materialism?

Before turning to specific examples, it should be noted that the Bible is often criticized because it was long regarded as completely true in a literal sense, whereas some of its passages were later found not to be accurate when interpreted literally, leading to at least partially metaphorical interpretations. Something similar can be said of materialism.

In the nineteenth century, materialism was a metaphysical doctrine asserting that only matter exists. However, for such a statement to have meaning, it is necessary to provide a definition of matter to which the statement applies.

In the nineteenth century, it was believed that the nature of matter was clear. The concept of matter was defined in terms of

- occupying a portion of space, meaning that two bodies cannot occupy the same region of space at the same time, from which the notions of
 - the position of a body in space,
 - the motion of a body through space over timeare derived;
- mass, which refers to
 - inertial properties, that is, a body's resistance to changes in its motion through space over time,

- gravitational properties, that is, a body’s influence on the motion of other bodies in its surroundings.

Under such a definition of matter, materialism meant that everything that exists has size and position in space, resists changes in its state of motion, and attracts everything else in its surroundings.

However, twentieth-century physics discovered that the photon has no mass (that is, it neither resists changes in its motion nor attracts other particles in its surroundings), and that quarks have no spatial extent but are instead treated as point-like objects. This means that the materialist assumption described above was not correct, and materialists, under the pressure of developments in physics, redefined their position.

However, this concerns a naive understanding of materialism. In this text, we shall define materialism as the assumption that only nature exists, where nature is understood in the sense in which it has been defined here. In other words, we define materialism as the assumption that everything that exists is measurable in principle and is subject to the Church–Turing–Deutsch thesis.

Similarly, we define reductionism as the assumption that the human mind is exclusively a matter of natural phenomena. In other words, reductionism is the assumption that the human mind is exclusively a matter of phenomena that are measurable in principle and are subject to the Church–Turing–Deutsch thesis. Of course, the materialist assumption implies the reductionist one.

Sir Roger Penrose, recipient of the Nobel Prize in Physics and of major international distinctions in mathematics, argues in his book “The Emperor’s New Mind” precisely for the negation of the reductionist assumption in the form stated above. The book received the Royal Society Science Book Prize.

9.1 The Success of Rationalists

Since rationalism tells us what we ought to do in the manner described above, the question naturally arises as to why rationalists are not the most successful people in the world.

Of course, the application of the methodology of the natural sciences to the study of natural phenomena is justified, and scientific theories describe nature at least approximately.

However, if matter is all that exists and if it is subject to the Church–Turing–Deutsch thesis, then this constitutes the entirety of reality. In that case, scientific theories describe reality itself, which is no longer limited merely to natural phenomena, because under the materialist assumption there are no other kinds of phenomena. Consequently, scientific theories describe reality as a whole, at least approximately.

Deviations between scientific theories and natural phenomena are attributed to ignorance, and ignorance is well modeled by randomness, that is, by probability theory.

However, this would provide no advantage to people who do not act rationally but instead behave in some other manner, because objective randomness applies equally to everyone.

The only explanation for the fact that rationalists are not the most successful people is that there exists something that is true, inaccessible to methodologies concerned exclusively with phenomena that are subject to the Church–Turing–Deutsch thesis, yet accessible by some other means and relevant to human life.

9.2 Qualia

When someone pricks us with a needle, we feel pain and say “ouch”. It is also possible to build a robot equipped with a sensor that detects a needle prick and causes the robot to produce the sound “ouch” when it is pricked. However, are these two situations really the same?

When a human being is pricked with a needle, the person not only exclaims “ouch” but also suffers. We do not ordinarily regard the robot that produces the sound “ouch” when pricked as having suffered. Qualia are defined as the difference between these two cases. A human being possesses qualia, that is, subjective inner experiences, whereas a robot that is entirely material does not. This suggests the existence of something nonmaterial.

9.3 The Measurement Problem in Physics

In quantum physical theories, the state of a physical system may be either reduced or unreduced. Initially, it is unreduced, and under certain conditions a reduction may occur, causing the state to become reduced.

In an unreduced state, a particle is located in all possible positions to some extent, according to a probability distribution. The same applies to the state of a physical system as a whole. In a reduced state, all possibilities except one have been eliminated.

However, the question naturally arises: under what conditions does reduction occur? The contemporary view is that there is no known set of purely physical conditions under which reduction necessarily occurs. Whatever physical processes take place, reduction need not occur.

It is commonly said that reduction occurs during a measurement, but this immediately raises the question of what a measurement actually is. What set of physical conditions is it that we call a measurement? This problem remains unsolved in physics to this day. Some of the greatest physicists in history have

stated that they do not know what a measurement is, or that if someone believes they have solved the measurement problem, then they have not understood the measurement problem.

There are several interpretations. One of them is that reduction occurs when a result is observed by consciousness, regarded as a nonphysical category. This is the interpretation physicists use in practice. This does not mean that the interpretation corresponds to reality, but only that it is sufficiently effective for doing physics. Another interpretation is the many-worlds interpretation, according to which there exist infinitely many universes, and every possible outcome occurs in one of them.

9.4 The Philosophy of Mathematics

There are views according to which mathematical realism is the only remaining viable option. If a mathematical universe exists, it must be far larger and richer than the material world, which means that the material world cannot be all that exists. In particular, if we define the material world in terms of the Church–Turing–Deutsch thesis, which mathematical laws do not satisfy, then the mathematical universe is more complex and richer than the material world.

Modern mathematics operates both with concepts that have at least an approximate realization in the material world and with concepts that have no approximate realization in the material world whatsoever. We shall refer to the latter as abstract concepts. Such concepts were not present in mathematics before the nineteenth century. More precisely, they first appear in mathematics with Galois theory.

If such concepts have no approximate realization in the material world, where are they realized, if anywhere? The standard foundation of mathematics, namely *ZFC*, has the following properties:

- Its axioms are very simple and intuitive.
- No one has succeeded in deriving a contradiction from these axioms, despite attempts by great mathematical minds.
- There exist much stronger systems (see the so-called “large cardinals”) that form a hierarchy in the following sense:
 - At the bottom of the hierarchy is *ZFC*.
 - Every system in the hierarchy except the first implies the consistency of all systems preceding it.

There are roughly twenty levels in this hierarchy. Thus, if we are not clever enough to derive a contradiction from *ZFC* even though one exists, then it should become increasingly easy to derive a contradiction as we move higher in this hierarchy, since the hierarchy is consistency-relevant. Nevertheless, no one has succeeded in deriving a contradiction from any of these stronger systems, despite attempts by great mathematical minds.

- The axioms imply the existence of objects that have no approximate realization in the material world.
- Concepts that have no approximate realization in the material world possess properties whose consequences do have at least an approximate realization.
- The axioms have so many consequences that they are applicable in almost every area of science and technology.
- There are no observations indicating that consequences of the axioms that do have at least an approximate realization fail in the material world.
- This axiomatic system that serves as a foundation for mathematics has remained unchanged since 1932 and still “holds up” in the sense described by the preceding points.

9.5 Reductionism and Free Will

Reductionism is also in conflict with the existence of free will. For an explanation of this issue, we recommend a video by Sabine Hossenfelder. Briefly stated, if the human mind is reducible to natural processes in the sense of being measurable in principle and subject to the Church–Turing–Deutsch thesis, then human behavior is a matter of mindless randomness rather than free will.

The video discusses the Free Will Theorem, according to which certain very general physical assumptions imply that if a human being possesses free will, then at least one elementary particle involved in the constitution of that human being must possess free will as well.

In other words, we cannot simultaneously accept both reductionism and the existence of free will. The nonexistence of free will also presents a problem for ethics.

10 Extraordinary Claims Require Extraordinary Evidence

The given definition of rationality implies what standards of evidence should look like. In order to obtain the most useful worldview, we should strive for a worldview that provides the most useful answers to the greatest number of questions while at the same time being as easy to apply as possible.

From the requirement that a worldview provide answers to as many questions as possible, we conclude that topics should not be treated as taboo. From the requirement that a worldview be as easy to apply as possible, we conclude that it should be as simple as possible. However, a worldview is more difficult to use if it is counterintuitive. Our intuitions are formed on the basis of experience, so this ultimately amounts to minimizing conflicts with intuition.

Nevertheless, people often speak of the objective simplicity of an explanation. This raises the question of how the simplicity of an explanation should be measured. Although no precise measure of explanatory complexity has been defined, in most cases it is clear which of two explanations is simpler, though this is not always the case.

The question of the universality of the principle of simplicity also arises. Reductionism is in conflict with the existence of free will. If reductionism is a simpler explanation than a worldview that includes free will, should reductionism be accepted? The nonexistence of free will creates difficulties for ethics, so a reductionist worldview is not necessarily the more useful one.

Let us now consider some examples illustrating how evidence may be viewed as a means of minimizing conflicts with intuition.

Special relativity is a highly counterintuitive theory. Nevertheless, its predictions, including some very counterintuitive ones, are supported by evidence. This means that denying any of these counterintuitive claims requires denying at least some part of every piece of supporting evidence. However, if we were to do so, we would create a much greater conflict with our intuitions than the counterintuitive claim of special relativity that we sought to reject.

Mathematics deals with concepts that are abstract in the sense that their interpretation is almost entirely arbitrary. The only restriction is that we must interpret constant symbols as entities that we call individuals, relation symbols as relations among individuals, and operation symbols as operations on individuals. We cannot interpret an operation symbol as something that is not an operation on those individuals.

Moreover, mathematics speaks only about the relationships among such concepts, and these relationships are themselves abstract mathematical objects in the same sense. The truth value of a mathematical statement does not exist indepen-

dently of a choice of interpretation of the mathematical concepts involved, except in special cases where the truth value is the same in every interpretation. Such abstract statements that are true in every interpretation are called logical laws.

The only thing that is indisputable in mathematics is that a statement called a theorem of an axiomatic theory is a logical consequence of the axioms of that theory; that is, there exists no interpretation in which all the axioms are true while the theorem is false. This is also the definition of a theorem.

Mathematicians prove that a statement is a theorem of a given axiomatic theory by constructing a logical bridge between the axioms and that statement. This bridge consists of a sequence of steps, each of which is an application of a basic logical rule or logical axiom, both of which are supported by very strong intuitions.

Mathematics contains many highly counterintuitive theorems. If we were to deny any such theorem of a theory for which a mathematical proof is available, we would have to reject at least one logical axiom or one basic logical rule. Doing so would produce a far greater conflict with our intuitions than the counterintuitive theorem itself. Mathematics also contains axioms that serve as foundations for the subject. These are chosen so as to provide the most useful and intuitive mathematics possible, and they are extended in accordance with that goal.

By an extraordinary claim we mean a claim that is either counterintuitive or has many consequences, so that drawing conclusions from it may lead to conflicts with intuition. If, on the basis of available knowledge, we judge the probability that a claim is true to be low, this too is a form of counterintuitiveness.

From this we arrive at the conclusion that extraordinary claims must be supported by extraordinary evidence, such that whenever the claim leads to counterintuitive consequences, the denial of the supporting evidence would produce an even greater conflict with intuition than acceptance of the extraordinary claim itself.

10.1 The Entropy of Our Big Bang

If we were to find seven marbles arranged as in Figure 10.1, we would assume that someone had thrown them or dropped them accidentally, because such an arrangement is not distinguished in any way; it is random and completely arbitrary. However, if we were to find seven marbles arranged as in Figure 10.1, we would assume that someone had deliberately arranged them. Let us consider this second situation.

This would not guarantee that someone actually arranged the marbles. Perhaps the marbles attract one another in some way while being subject to a constraint preventing them from approaching closer than a certain distance, causing them to settle into an optimal configuration. For example, they might be located at the centers of transparent balloons that we cannot see, as shown in Figure 10.1.

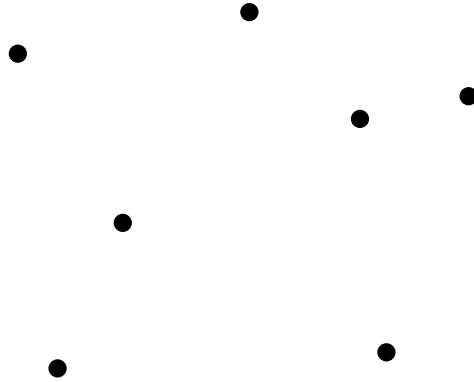


Figure 1: Randomly arranged points

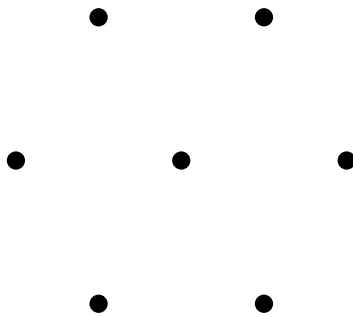


Figure 2: Regularly arranged points

But what if the observed regularity is not a consequence of natural laws? If we were to throw the marbles randomly, the probability that they would occupy exactly the configuration shown would be zero. However, there is a positive probability that they would be arranged approximately in that way. Moreover, the greater the required precision, the smaller the probability.

An example from nature is provided by the shapes of planets and stars, which in an overwhelming number of cases are spherical to a very high degree of accuracy,

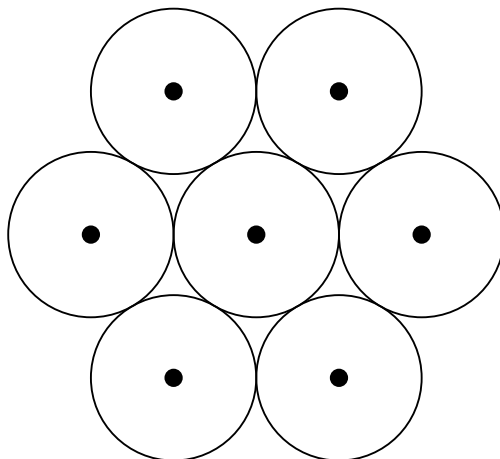


Figure 3: Invisible balloons

as though some powerful being had carefully sculpted them. In reality, this shape is a consequence of the natural law of gravitation.

If we were to melt a mobile phone into a homogeneous mass, thoroughly mix that mass, and then randomly rearrange the atoms from which the phone was made within the volume previously occupied by the phone, the probability of obtaining a functioning mobile phone would be negligible.

This is precisely why a mobile phone could not have arisen spontaneously. It could only have been produced by sufficiently intelligent beings such as humans. Let us therefore consider how humans create such things.

If we need to move a bucket from one place on a table to another, we can do so easily, because if the bucket were moved randomly there would still be a substantial probability that it would end up in the desired region of the table.

However, threading a needle is much more difficult, because if the thread were moved randomly, the probability of success would be very small. Nevertheless, it is not excessively small. It is large enough that, if we proceed carefully and make enough attempts, we eventually succeed.

Such an achievement is accomplished through a sequence of steps, each of which is simple in the sense that, if events within that step were left to chance, there would still be a significant probability that the step would occur in the desired way. Yet the probability that every step in a long sequence of steps would spontaneously occur in the desired manner is effectively zero. We use knowledge of what must be done at each stage, and thereby succeed in producing something that could not arise spontaneously.

According to Roger Penrose, astrophysicists have determined that our Big Bang

originated in an extremely fine-tuned state. In other words, a statistical miracle has been observed—something whose spontaneous occurrence is exceedingly improbable. There are several possible explanations for this.

The first explanation is that if an outcome of an experiment has even a slightly positive probability, then given infinitely many trials that outcome will occur. Thus, if there are infinitely many Big Bangs, and if the observed statistical miracle has positive probability—that is, if it is not ruled out by some law—then it is to be expected that there exist Big Bangs in which it occurs.

However, this means that such a statistical miracle appears only very rarely among Big Bangs. One may then ask why our Big Bang is one of these extremely rare cases. This would imply that such a statistical miracle is necessary for cosmic evolution to produce intelligent beings capable of observing and measuring it, such as ourselves. In that case, we simply could not have arisen in a Big Bang in which it did not occur.

Nevertheless, this requires an explanation of how the observed fine-tuning is related to the possibility of the emergence of intelligent beings such as ourselves. No such explanation has been provided.

The second explanation is that the fine-tuning is merely apparent, and that the observed state is actually the consequence of natural laws, just as in the example of marbles located at the centers of invisible spheres. This would mean that, for reasons connected with the generation of Big Bangs, only such Big Bangs arise, or at least arise with very high probability, so that our own Big Bang is not special after all. However, the mechanism by which Big Bangs originate is not known.

Roger Penrose has proposed such a type of explanation through the introduction of dark matter and a model of Big Bang formation according to which the universe resulting from one Big Bang expands and cools for an extremely long time until, in a certain manner, a new Big Bang emerges from it. According to this model, our Big Bang originated from a previous one, which in turn originated from an earlier one, and so on without end into the past.

The third explanation is intelligent design, namely that our Big Bang was designed by some intelligent being. In that case, the being would have to be extraordinarily powerful. The measured results correspond to the following scenario: suppose creation proceeded in a sequence of steps such that

- at each step there are 10 possible outcomes, all equally likely under spontaneous evolution,
- at each step only one of those 10 outcomes is correct,
- the being knew at every step which outcome was correct and always selected it,

then the number of steps in the creation process would have to be approximately 10^{123} .

Religious believers, of course, conceive of God as a being fully capable of accomplishing such a task. However, from a rationalist perspective, the difficulty with this explanation is that the claim that such a powerful being exists is itself an extremely strong claim, and therefore requires extraordinary evidence. No evidence for the existence of such a being is available. This is the reason Roger Penrose rejects this possibility.

At this point, we shall explore the possibility that the assumption of the existence of such a being may not be as implausible as it initially appears.

10.2 Is the Assumption That God Exists Implausible?

God is usually conceived as an omnipotent and omniscient being. However, unlimited power is not necessary for a being to be called God. It is sufficient that such a being created our Big Bang and is powerful enough to manipulate us and our Big Bang while remaining beyond our ability to affect it in any way.

Let us imagine a broader universe than our Big Bang, in which our Big Bang is merely a part. If we reject reductionism, then our Big Bang cannot be all that exists or all that is connected with us.

This broader reality would have to be much larger, wider, and richer than our Big Bang. The question then arises whether there exist intelligent beings within that broader reality that are capable of creating Big Bangs like ours and powerful enough to manipulate us and our Big Bang.

The fact that conditions suitable for intelligent life exist within our Big Bang does not guarantee that corresponding conditions are satisfied in that broader universe. Perhaps some prerequisite for the existence of intelligent beings is absent there, despite the fact that it is much larger and richer than our own universe, which satisfies all the necessary conditions.

In mathematics it is known that knots are possible only in three-dimensional space. Although spaces of dimension greater than three may be regarded as broader and richer than three-dimensional space, they possess enough degrees of freedom that everything can be untangled, making knots impossible. Thus, as a universe becomes broader and richer, certain possibilities may disappear.

Let us consider the case in which there exist intelligent beings in that broader reality that are capable of creating things within their universe.

In order for us humans to exert a significant influence on our environment, we must possess sufficient complexity and size relative to the structure of our Big Bang, that is, relative to the organization of matter, the scale of molecules, and the complexity of the laws governing our universe.

If that broader universe in which Big Bangs occur possesses a much more complex internal structure, then the power of beings within it should also be much greater. In that context, the number 10^{123} , which is enormous from our perspective, might in fact be small.

If that broader universe includes the mathematical universe, then it must be extraordinarily complex, since the internal structure of the mathematical universe is itself extraordinarily complex.

However, the universe in which Big Bangs occur need not include the mathematical universe. The mathematical universe could itself be part of some even broader universe.

Nevertheless, such a broader universe would have to be much richer and more complex, and therefore the beings capable of creating within it would also have to be much more powerful. If all types of particles that can mathematically exist are realized within our Big Bang (which is what observations suggest), then why should this not also be the case in that broader universe?

This should not be confused with the simplicity of the description of our Big Bang. By comparison, all algorithms can be reduced to compositions of a few simple rules, yet the universe of algorithms is so complex that no method exists that can, in general, determine whether an arbitrary algorithm halts.

11 The Meaning of Our Universe

Imagine that intelligent beings live on molecules. Their scientists would observe natural laws governing the arrangement of electrons in atomic orbitals, as well as laws governing valence. Their astronomers would then observe other molecules moving around them.

If those molecules happened to be part of an exploding bomb, those beings would observe the expansion and cooling of their universe. However, they would be unable to understand its purpose, which might be that someone was at war with someone else and had therefore dropped a bomb. The purpose would exist, but it would remain inaccessible to them.

What is essential is only that the relationship between consciousness and a robot satisfying the same materialist conditions as a human body be established in the same manner as the relationship between consciousness and the material component of a human being.

12 Humans and Machines

There is a widespread belief that certain religious assumptions imply that human beings can do things that machines will never be able to do. This view is usually supported by the claim that religious assumptions imply that a human being is not purely material in nature, and even that the human mind cannot be reduced solely to materialistic processes, whereas a machine is something entirely materialistic. Let us introduce the assumption that a soul exists and that it belongs to a transcendent realm.

A human being certainly has a material component, which we call the body and which includes the brain, while the processes occurring within the brain are entirely materialistic. At the same time, the existence of a soul is assumed, and this soul is transcendent and somehow interacts with the brain.

For example, when we perceive something, our senses first detect it. The information is then transmitted to the brain, where certain processes take place. This part is entirely materialistic. The soul then reads these brain states, and a corresponding qualia is produced.

Conversely, when we decide to move our hand, that decision ultimately results in the movement of the hand, which is entirely material, so the movement of the hand is a movement of matter. In this case, the soul first produces certain brain states, after which a completely materialistic process leads to the movement of the hand.

However, in this way the soul can interact only with material objects that satisfy certain materialistic conditions (of the kind satisfied by the brain). Among these conditions could be a sufficient degree of indeterminacy in the state of such an object.

The operation of the brain is highly deterministic because its primary function is to maintain vital functions. If this function were to fail for some period due to insufficient reliability, death would result. Therefore, the brain must be a highly reliable device.

However, its operation is not completely deterministic. The degree of indeterminacy in the brain's operation is not large, but it is sufficient for the soul to make choices that enable us to do whatever we decide within the material world.

We may suppose that there exists a transcendent ocean of souls that interact with material objects satisfying certain materialistic conditions, conditions that the human body satisfies. However, if we construct a robot that is entirely material and that satisfies those same materialistic conditions, then a soul would inhabit it in the same way, enabling it to possess all the properties of a human being.

If the assignment of souls to objects were determined by God, then the above conclusion would follow provided that God's will is not discriminatory in favor of humans, that is, provided that it applies equally to humans and machines.

The description of the soul as something that exists separately from the body and is somehow joined to the body corresponds to religions that include reincarnation, whereas Christian conceptions are different.

However, the above description is flexible enough to accommodate such modifications. What is actually essential is only that the relationship between consciousness and a robot that satisfies the same materialistic conditions as a human body be established in the same way as the relationship between consciousness and the material component of a human being.